

《复变函数与积分变换》 2021年答案

一、1. $z = -1 - \sqrt{3}i = 2e^{-\frac{2\pi}{3}}$

$\therefore \sqrt[n]{z} = \sqrt[n]{2} \cdot e^{\frac{-\frac{2\pi}{3} + 2k\pi}{n}}$

$= \sqrt[n]{2} e^{(-\frac{\pi}{n} + \frac{2k\pi}{n})i}, k=0,1,2,3,4,5$



或 $= \sqrt[n]{2} \left[\cos(-\frac{\pi}{n} + \frac{2k\pi}{n}) + i \sin(-\frac{\pi}{n} + \frac{2k\pi}{n}) \right], k=0,1,2,3,4,5$

2. $(2-4i)^i = e^{i \ln(2-4i)} = e^{i [\ln(2\sqrt{5}) + i(-\arctan 2 + 2k\pi)]} = 2\sqrt{5} e^{-\arctan 2}$

$= e^{\arctan 2 - 2k\pi} [\cos(\ln 2\sqrt{5}) + i \sin(\ln 2\sqrt{5})], k \in \mathbb{Z}$



3. $\int_C (x-2y^2i) dz$, 其中C为沿曲线 $y=\sqrt{x}$ 从 $1+2i$ 到 $4+4i$ 的一段弧。

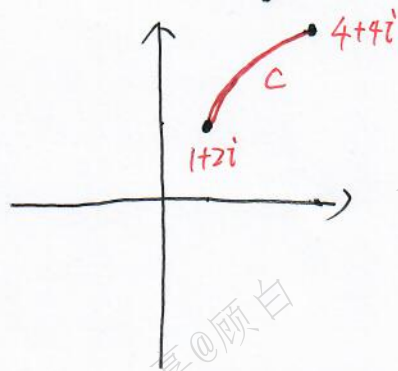
解: 曲线C的参数方程: $\begin{cases} x = t^2 \\ y = 2t, 1 \leq t \leq 2 \end{cases}$

$\Rightarrow z(t) = t^2 + i \cdot 2t, z'(t) = 2t + 2i$

$\therefore \int_C (x-2y^2i) dz$

$= \int_1^2 (t^2 - 2 \cdot (2t)^2 i) (2t + 2i) dt$

$= \int_1^2 2(1-8i)(t^3 + it^3) dt = 2(1-8i) \times \left(\frac{t^4}{4} + \frac{it^3}{3} \right) \Big|_1^2 = 2(1-8i) \times \left(\frac{15}{4} + \frac{7i}{3} \right) = \frac{269}{6} - \frac{166i}{3}$



4. $f(z) = \frac{1}{z+3}$, 求 $f^{(10)}(2)$.

解: $f^{(10)}(2) = \frac{10!}{2\pi i} \oint_C \frac{f(z)}{(z-2)^{11}} dz$, 其中C为包含 $z=2$ 的闭曲线

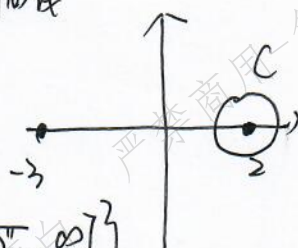
$= \frac{10!}{2\pi i} \oint_{|z-2|=\gamma} \frac{1}{(z+3)(z-2)^{11}} dz$

留数+第二定理

$= -\frac{10!}{2\pi i} \times 2\pi i \left\{ \text{Res} \left[\frac{1}{(z+3)(z-2)^{11}}, -3 \right] + \text{Res} \left[\frac{1}{(z+3)(z-2)^{11}}, \infty \right] \right\}$

$= -10! \times \left\{ \frac{1}{(-5)^{11}} - \text{Res} \left[\frac{1}{(\frac{1}{z} + \frac{1}{z-2})^{11}} \cdot \frac{1}{z^2}, 0 \right] \right\}$

$= \frac{10!}{5^{11}}$



$$5. \int_0^{\infty} t \sin t e^{-t} dt$$

$$= \mathcal{L}[t \sin t] \Big|_{s=1} = \frac{2s}{(s^2+1)^2} \Big|_{s=1} = \frac{1}{2}$$

$$\text{注: } \mathcal{L}[t \sin t] = -(\mathcal{L}[\sin t])' = -\left(\frac{1}{s^2+1}\right)' = \frac{2s}{(s^2+1)^2}$$

$$\begin{aligned} \rightarrow \int_0^{\infty} t \sin t dt &= \int_0^{\infty} t \sin t \cdot e^{0t} dt \\ &= \mathcal{L}[t \sin t] \Big|_{s=0} \\ &= \frac{2s}{(s^2+1)^2} \Big|_{s=0} = 0 \end{aligned}$$

$$二. f(z) = \frac{(\cos \pi z + 1)^2 (z+i)}{(z^2-1)(z^2+1)} \quad \text{在扩充复平面内的孤立奇点并分类}$$

$$\text{解: 令 } P(z) = (\cos \pi z + 1)^2 (z+i), \quad Q(z) = (z^2-1)(z^2+1)$$

$$\text{由 } Q(z)=0 \Rightarrow z=\pm 1, z=\pm i, \quad \underline{z_{\infty}=\infty}$$

$$\text{即 } P(z) \text{ 的零点有: } z_k = 2k+1, k \in \mathbb{Z}, z=-i$$

$$\textcircled{1} z=\pm 1, \text{ 为 } P(z) \text{ 的 } 2 \text{ 级零点, } Q(z) \text{ 的 } 1 \text{ 级零点} \Rightarrow f(z) \text{ 的可去奇点}$$

$$\textcircled{2} z=-i, \text{ 为 } P(z) \text{ 的 } 1 \text{ 级零点, } Q(z) \text{ 的 } 1 \text{ 级零点} \Rightarrow \text{可去奇点}$$

$$\textcircled{3} z=i, \text{ 不是 } P(z) \text{ 的零点, 是 } Q(z) \text{ 的 } 1 \text{ 级零点} \Rightarrow 1 \text{ 级极点}$$

$$\textcircled{4} z=\infty, \text{ 令 } z=\frac{1}{t} \quad f\left(\frac{1}{t}\right) = \frac{[\cos(\frac{\pi}{t})+1]^2 (\frac{1}{t}+i)}{[\frac{1}{t^2}-1][\frac{1}{t^2}+1]}$$

$$\text{出现 } \cos(\frac{\pi}{t}) \text{ 说明 } t=0 \text{ 为 } f\left(\frac{1}{t}\right) \text{ 的本性奇点}$$

$$\Rightarrow z=\infty \text{ 为 } f(z) \text{ 的本性奇点}$$

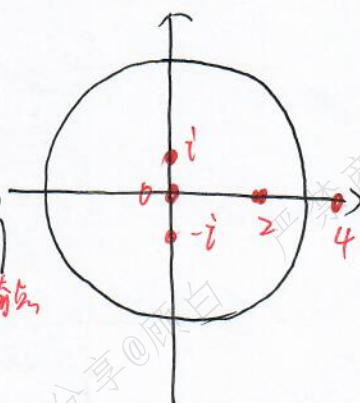
$$三. \oint_{|z|=3} \frac{1}{z(z^2+1)^3(z-2)^2(z-4)} dz \quad \xrightarrow{f(z)}$$

$$= -2\pi i \times \{ \text{Res}[f(z), 4] + \text{Res}[f(z), \infty] \}$$

$$= -2\pi i \times \left\{ \frac{1}{4 \times 1^3 \times 4} - \text{Res}\left[\frac{1}{z(z^2+1)^3(z-2)^2(z-4)} \cdot \frac{1}{z^2}, 0\right] \right\}$$

可去奇点

$$= -2\pi i \times \frac{1}{16 \times 1^3} = -\frac{\pi i}{8 \times 1^3}$$



$$2. \int_0^{+\infty} \frac{\sin t}{t} e^{-t} dt$$

$$= \mathcal{L} \left[\frac{\sin t}{t} \right] \Big|_{s=1}$$

$$= \int_s^{\infty} \mathcal{L}[\sin t] ds \Big|_{s=1}$$

$$= \int_1^{\infty} \frac{1}{s^2+1} ds$$

$$= \arctan s \Big|_1^{\infty} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$3. \int_{-\infty}^{+\infty} \frac{\sin x}{x^2+2x+4} dx$$

$$= \operatorname{Im} \left[\int_{-\infty}^{+\infty} \frac{e^{ix}}{x^2+2x+4} dx \right]$$

$$= \operatorname{Im} \left\{ 2\pi i \times \operatorname{Res} \left[\frac{e^{iz}}{z^2+2z+4}, -1+\sqrt{3}i \right] \right\}$$

$$= \operatorname{Im} \left\{ 2\pi i \times \frac{e^{i(-1+\sqrt{3}i)}}{2\sqrt{3}i} \right\}$$

$$= -\frac{\pi}{\sqrt{3}e^{\sqrt{3}}} \sin 1$$

$$4. \int_{-\infty}^{+\infty} \frac{x^2}{(4+x^2)(1+x^2)} dx \quad (\text{也可以取一极}) \quad \text{题目抄错了} \rightarrow$$

$$= 2\pi i \times \left\{ \operatorname{Res} \left[\frac{z^2}{(z^2+4)(z^2+1)}, 2i \right] + \operatorname{Res} \left[\frac{z^2}{(z^2+4)(z^2+1)}, i \right] \right\}$$

$$= 2\pi i \times \left\{ \frac{-4}{4i \times (-3)} + \frac{-1}{3 \times 2i} \right\} = 2\pi i \times \left(\frac{1}{3i} - \frac{1}{6i} \right) = \frac{\pi}{3}$$

$$\begin{aligned} 4. \int_{-\infty}^{+\infty} \frac{\cos x}{4+x^2} dx \\ &= \operatorname{Re} \left[\int_{-\infty}^{+\infty} \frac{e^{ix}}{4+x^2} dx \right] \\ &= \operatorname{Re} \left\{ 2\pi i \times \operatorname{Res} \left[\frac{e^{iz}}{4+z^2}, 2i \right] \right\} \\ &= \operatorname{Re} \left\{ 2\pi i \times \frac{e^{-2}}{4i} \right\} \\ &= \frac{\pi}{2e^2} \end{aligned}$$

四. 1. $w = z^2$ 将 $y = x$ 映成 w 平面何种曲线?

$$\text{解: } w = z^2 = (x+iy)^2 = x^2 - y^2 + 2xyi \Rightarrow \begin{cases} u = x^2 - y^2 \\ v = 2xy \end{cases}$$

$$\text{将 } y=x \text{ 代入} \Rightarrow \begin{cases} u = x^2 - 4x^2 = -3x^2 \\ v = 2x \cdot x = 4x^2 \end{cases}$$

$$\Rightarrow \frac{u}{v} = -\frac{3}{4} \text{ 为直线}$$

$$v = -\frac{4}{3}u$$



2. $f(z) = \frac{1}{z^2+1}$ 在圆环域 $1 < |z - \underbrace{(1+i)}_{z_0}| < \sqrt{5}$ 内展开成洛朗级数

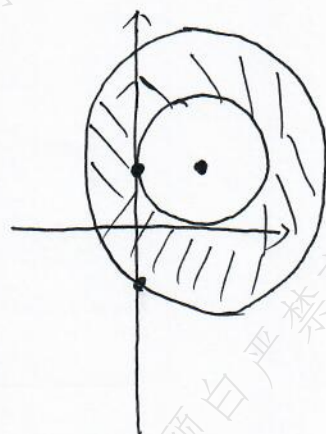
$$\text{解: } f(z) = \frac{1}{2i} \left(\frac{1}{z-i} - \frac{1}{z+i} \right)$$

$$= \frac{1}{2i} \left(\frac{1}{[z-(1+i)]+1} - \frac{1}{[z-(1+i)]+(1+2i)} \right)$$

$$= \frac{1}{2i} \left[\frac{1}{z-(1+i)} \cdot \frac{1}{1 + \frac{1}{z-(1+i)}} - \frac{1}{1+2i} \cdot \frac{1}{1 + \frac{z-(1+i)}{1+2i}} \right]$$

$$= \frac{1}{2i} \left[\frac{1}{z-(1+i)} \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{z-(1+i)} \right)^n - \frac{1}{1+2i} \sum_{n=0}^{\infty} (-1)^n \left(\frac{z-(1+i)}{1+2i} \right)^n \right]$$

$$= \frac{1}{2i} \left[\sum_{n=0}^{\infty} (-1)^n \frac{1}{[z-(1+i)]^{n+1}} - \frac{1}{1+2i} \sum_{n=0}^{\infty} (-1)^n \frac{[z-(1+i)]^n}{(1+2i)^{n+1}} \right]$$



五. $u = x^2 - y^2 + 2xy - x + y$, 证明 u 为调和函数, 并求 v , 使得 $f(z) = u + iv$ 为解析函数, 满足 $f(i) = 0$

解: i) $\frac{\partial u}{\partial x} = 2x + 2y - 1$, $\frac{\partial u}{\partial y} = -2y + 2x + 1$

$\frac{\partial^2 u}{\partial x^2} = 2$, $\frac{\partial^2 u}{\partial y^2} = -2 \Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0 \Rightarrow u$ 为调和函数

ii) 由 C-R 方程:

$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = 2x + 2y - 1$ ①

$\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = 2y - 2x - 1$ ②

对①关于 y 积分 $\Rightarrow v(x, y) = \int (2x + 2y - 1) dy = 2xy + y^2 - y + \phi(x)$

$\therefore \frac{\partial v}{\partial x} = 2y + \phi'(x)$ 与②比较

$\Rightarrow \phi'(x) = -2x - 1 \Rightarrow \phi(x) = \int (-2x - 1) dx = -x^2 - x + C$

$\therefore v(x, y) = 2xy + y^2 - y - x^2 - x + C$

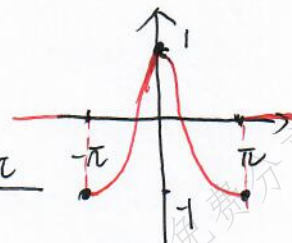
$\Rightarrow f(z) = u + iv = x^2 - y^2 + 2xy - x + y + i(2xy + y^2 - y - x^2 - x + C)$

$f(i) = 0 + iC = 0 \Rightarrow C = 0 \therefore f(z) = \underline{(C=0)}$

7. 求 $f(t) = \begin{cases} \cos t + \sin t, & |t| \leq \pi \\ 0, & |t| > \pi \end{cases}$ 的傅氏变换, 并求 $\int_0^{\infty} \frac{w \sin w \pi \cos w t + \sin w \pi \sin w t}{1 - w^2} dw$ 的值

解: i) $F(w) = \int_{-\infty}^{\infty} f(t) e^{-iwt} dt = \int_{-\pi}^{\pi} (\cos t + \sin t) e^{-iwt} dt$

$= \int_{-\pi}^{\pi} (\cos t \cos wt - i \sin t \sin wt) dt = \frac{2w \sin w \pi - 2i \sin w \pi}{1 - w^2}$



ii) $f^*(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(w) e^{iwt} dw = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{w \sin w \pi - i \sin w \pi}{1 - w^2} \cdot (\cos wt + i \sin wt) dw$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{w \sin w \pi \cos wt + \sin w \pi \sin wt}{1 - w^2} dw = \begin{cases} 0, & |t| > \pi \\ \cos t + \sin t, & |t| < \pi \\ \frac{1+0}{2} = \frac{1}{2}, & |t| = \pi \end{cases}$

$\Rightarrow \int_0^{\infty} \frac{w \sin w \pi \cos w t + \sin w \pi \sin w t}{1 - w^2} dw = \begin{cases} 0, & |t| > \pi \\ \frac{\pi}{2} (\cos t + \sin t), & |t| < \pi \\ -\frac{\pi}{4}, & |t| = \pi \end{cases}$

$$t. \quad \begin{cases} y'' + y' - 6y = e^{2t} \\ y(0) = y'(0) = 0. \end{cases}$$

解: 令 $\mathcal{L}[y(t)] = Y(s)$.

$$|21| \quad \mathcal{L}[y'' + y' - 6y] = \mathcal{L}[e^{2t}]$$

$$\Rightarrow s^2 Y(s) - \overset{=0}{s y(0)} - \overset{=0}{y'(0)} + s Y(s) - y(0) - 6 Y(s) = \frac{1}{s-2}$$

$$\Rightarrow (s^2 + s - 6) Y(s) = \frac{1}{s-2} \Rightarrow Y(s) = \frac{1}{(s-2)^2(s+3)}$$

$$\therefore y(t) = \mathcal{L}^{-1}[Y(s)] = \mathcal{L}^{-1}\left[\frac{1}{(s-2)^2(s+3)}\right]$$

$$= \text{Res}\left[\frac{e^{st}}{(s-2)^2(s+3)}, 2\right] + \text{Res}\left[\frac{e^{st}}{(s-2)^2(s+3)}, -3\right]$$

$\downarrow$
二重极点 $\downarrow$
单极点

$$= \left(\frac{e^{st}}{s+3}\right)' \Big|_{s=2} + \frac{e^{st}}{(s-2)^2} \Big|_{s=-3}$$

$$= \frac{e^{-3t}}{25} + \frac{t e^{2t}}{5} - \frac{e^{2t}}{25}$$